

WHAT MAKES SAM'S CLUB'S SUPPLY CHAIN SUCCESSFUL IN CHINA? AN ANALYSIS OF SUPPLY CHAIN MANAGEMENT AS A SOURCE OF COMPETITIVE ADVANTAGE

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Abstract: Retail competition is increasingly a competition between supply chains rather than between individual products or stores. This paper analyzes the supply chain of Sam's Club in China and argues that its competitive advantage rests on four capabilities, namely a limited stock-keeping-unit (SKU) strategy, membership-driven demand visibility, centralized procurement, and data-driven supply chain management. Drawing on established concepts from operations and economics, including economies of scale, information sharing, and inventory theory, the paper explains how each capability lowers cost or reduces uncertainty and how the four capabilities reinforce one another. A limited assortment raises purchase volume per item and strengthens bargaining power. Membership converts transactions into demand information that improves forecasting. Centralized procurement realizes scale economies, and data-driven operations balance product availability against inventory cost. The paper also examines the main challenges, namely global sourcing risk, residual demand uncertainty, and the trade-off between efficiency and resilience. The analysis concludes that supply chain management is not merely a support function but a durable source of competitive advantage, and it notes that firms compete increasingly through the coordination of information, inventory, suppliers, and logistics.

Keywords: Supply chain management; Competitive advantage; Economies of scale; Demand forecasting; Retail; Membership model

1 INTRODUCTION

When most consumers think about Sam's Club, they usually focus on its membership model, imported products, or high-quality private-label goods. Many people believe that the company's success comes primarily from charging membership fees and offering premium products. A closer look at the business model suggests a different question, namely what allows Sam's Club to consistently provide popular products, maintain relatively competitive prices, and sustain high customer satisfaction in China's highly competitive retail market.

At first glance Sam's Club appears to be a retailer, yet its true competitive advantage lies deeper within its operations. Behind every product that customers purchase is a complex network of suppliers, warehouses, transportation systems, forecasting models, and inventory decisions. Modern business competition is increasingly not a competition between stores, products, or advertisements. Instead, it is often a competition between supply chains [1-2].

This paper argues that the success of Sam's Club in China is largely driven by four supply chain capabilities, namely a limited SKU strategy, membership-driven demand visibility, centralized procurement, and data-driven supply chain management. Together these capabilities allow the company to create value for customers while maintaining operational efficiency. The remainder of the paper explains why supply chains matter, analyzes each of the four capabilities in turn, discusses the main challenges and trade-offs, and concludes with the broader lesson that supply chain management can be a source of sustained competitive advantage.

2 WHY SUPPLY CHAINS MATTER

Most consumers rarely think about supply chains. When customers buy products from a supermarket, they usually focus on price, product quality, and the store experience. However, all of these experiences are shaped by supply chain decisions. If a retailer cannot accurately predict demand, popular products may frequently run out of stock. If procurement decisions are inefficient, product costs rise. If logistics operations fail, products may arrive late or become unavailable. Supply chain performance therefore directly affects customer satisfaction [3].

In this sense, companies do not simply sell products. They organize complex networks of suppliers, manufacturers, warehouses, distributors, and retailers to create value for consumers. This perspective motivates an analysis of Sam's Club from a supply chain point of view, because the visible retail experience is the output of many upstream decisions. Table 1 summarizes the four capabilities analyzed in this paper and the primary mechanism through which each creates value.

Table 1 Four Supply Chain Capabilities and Their Mechanisms

Capability	Primary mechanism
Limited SKU strategy	Higher volume per item, stronger bargaining power, lower cost
Membership demand visibility	Transactions become demand data that improve forecasting

Capability	Primary mechanism
Centralized procurement	Economies of scale lower average unit cost
Data-driven management	Balances availability against inventory cost

3 LIMITED SKU STRATEGY

One of the most distinctive features of Sam's Club is its limited product assortment. Traditional supermarkets often carry tens of thousands of stock-keeping units (SKUs), whereas Sam's Club intentionally offers a much smaller selection. At first this strategy appears counterintuitive, because most people assume that offering more products should attract more customers. From a supply chain perspective, however, reducing the number of SKUs creates significant advantages. The core logic can be summarized as a causal chain: Fewer SKUs → larger volume per product → stronger bargaining power → lower procurement cost → better customer value.

Because Sam's Club purchases larger quantities of fewer products, it can negotiate better prices with suppliers [4]. A smaller assortment also reduces operational complexity, because the company needs fewer suppliers, simpler forecasting, less inventory-management complexity, and more efficient replenishment. As a result, resources can be concentrated on maintaining product quality and availability. This strategy illustrates an important business lesson, namely that more choice does not always create more value. Sometimes limiting choice allows a company to operate more efficiently and deliver a better customer experience [5].

4 MEMBERSHIP AND DEMAND VISIBILITY

Many people assume that membership fees are simply an additional source of revenue. The more important value of the membership model, however, lies in the information it creates. Unlike a traditional supermarket, Sam's Club knows exactly who its customers are, and every purchase generates data. The company can observe purchasing frequency, product preferences, seasonal patterns, and customer loyalty, which together allow it to better understand future demand. The relationship can be represented as a chain: Membership data → demand forecasting → inventory planning → product availability → customer satisfaction.

One of the greatest challenges in supply chain management is uncertainty, because companies rarely know exactly what customers will buy in the future. This uncertainty produces well-documented problems such as the amplification of demand variability along the chain, often called the bullwhip effect [6]. Membership data reduces this uncertainty at its source. With better forecasts, Sam's Club can reduce stockouts, lower excess inventory, improve service levels, and reduce operating costs [7]. Viewed this way, membership is not only a pricing strategy but also an information system, and in modern business information is often one of the most valuable assets a firm can possess [8].

5 CENTRALIZED PROCUREMENT AND ECONOMIES OF SCALE

Another important characteristic of Sam's Club is its centralized procurement system. Instead of allowing individual stores to negotiate with suppliers independently, purchasing decisions are coordinated centrally, which creates significant economies of scale. The mechanism can be represented as a chain: Large aggregated demand → large purchase orders → lower unit cost → competitive prices.

Through centralized procurement, Sam's Club can negotiate lower prices, standardize quality requirements, simplify supplier management, and improve purchasing efficiency. Suppliers also often prefer to work with large buyers, because larger orders provide stable demand, so both sides benefit from long-term relationships. From an economic perspective, this reflects the principle of economies of scale, in which the average cost per unit falls as purchasing volume rises [9]. This provides a sustainable source of competitive advantage that smaller retailers may struggle to replicate, because scale advantages are difficult to imitate without comparable volume.

6 DATA-DRIVEN SUPPLY CHAIN MANAGEMENT

Modern supply chains increasingly rely on data rather than intuition. Sam's Club collects information from membership transactions, online orders, inventory systems, and sales records, and this information supports decision-making throughout the chain. The typical process can be represented as a chain: Customer demand → data collection → forecasting → inventory decisions → replenishment.

The goal is to ensure that products are available when customers need them while avoiding unnecessary inventory. This balance is important because inventory creates both opportunities and risks. Too little inventory leads to stockouts, and too much inventory leads to waste and higher holding costs, so data helps the company find the right balance [7]. Perhaps the most important insight is that modern supply chains compete increasingly through information. Historically, companies addressed uncertainty by holding more inventory. Today they increasingly address it through better data and forecasting, which lowers cost and improves service at the same time [8].

7 CHALLENGES AND TRADE-OFFS

Despite its strengths, the supply chain of Sam's Club faces several challenges that qualify the advantages described above.

7.1 Global Supply Chain Risk

Many products sold by Sam's Club are sourced internationally, which creates exposure to shipping disruptions, exchange-rate fluctuations, trade-policy changes, and geopolitical uncertainty. A disruption anywhere in the chain can affect product availability and cost, and long international lead times make recovery slower [10].

7.2 Demand Uncertainty

Although membership data improves forecasting, demand can never be predicted perfectly. Unexpected events, changing consumer preferences, and shifting economic conditions all affect purchasing behavior, so forecasting error remains a permanent feature of the problem rather than a temporary limitation.

7.3 Efficiency versus Resilience

Perhaps the most important challenge is the trade-off between efficiency and resilience. An efficient supply chain minimizes cost and inventory, while a resilient supply chain maintains flexibility and protection against disruption. The two objectives often conflict, because more inventory improves resilience but raises cost, whereas lower inventory improves efficiency but raises risk [11]. Managers must continuously balance these competing objectives. This trade-off is not unique to Sam's Club, because it represents a fundamental challenge faced by many firms, and the appropriate balance shifts with the level of disruption risk in the environment.

8 DESCRIPTIVE DATA ANALYSIS

8.1 Data Scope and Method

Tables 2-4 and Figures 1-3 use official Walmart China disclosures for FY2024, covering February 2023 through January 2024 [12-15]. The reporting scope includes Sam's Club and other Walmart China formats; these are not standalone Sam's Club China results. Growth rates compare each quarter with the corresponding prior-year period, on the constant-currency basis specified by Walmart. Comparable sales include eCommerce. This descriptive analysis provides business context rather than a causal test of supply chain efficiency.

8.2 Sales Growth

Table 2 Walmart China Sales Growth in FY2024

Quarter	Net sales YoY (%)	Comparable YoY (%)
Q1	28.3	25.5
Q2	21.7	17.2
Q3	25.3	18.6
Q4	11.3	6.6

Sources: China disclosures [12-15]. YoY = year on year.

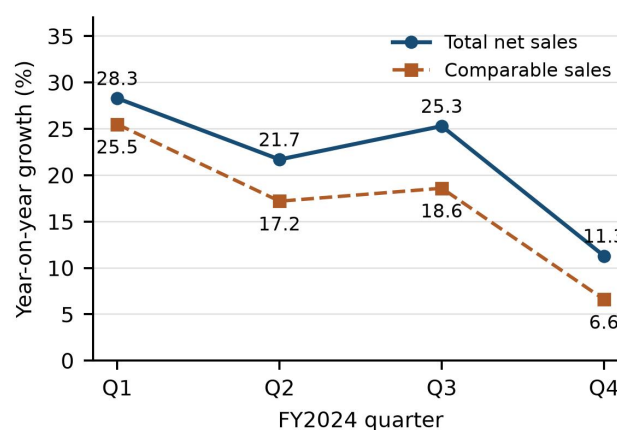


Figure 1 Walmart China Net and Comparable Sales Growth.
Sources: [12-15].

Both measures remained positive in all four quarters. Their variation indicates that replenishment plans should accommodate changing demand conditions. Walmart attributed part of the Q4 slowdown to the timing of Chinese New Year [15]; the pattern alone cannot establish deteriorating supply chain performance.

8.3 Digital Channel Mix

Table 3 Walmart China Digital Indicators in FY2024

Quarter	eCommerce YoY (%)	eCommerce share (%)
Q1	54	40
Q2	44	47
Q3	38	45
Q4	11	48

Sources: Share denotes the proportion of net sales classified as eCommerce [12-15].

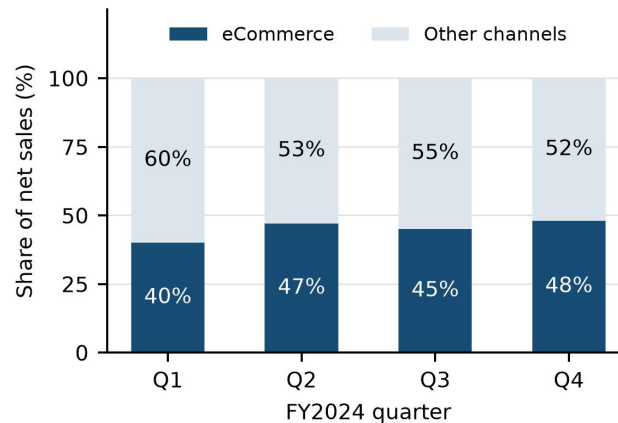


Figure 2 Walmart China Channel Mix

Sources: Other channels = 100% minus eCommerce share [12-15].

Digital penetration rose from 40% to 48% between Q1 and Q4, with a Q3 dip [12-15]. The changing mix makes coordination of digital orders and store inventory relevant to the case argument. These disclosures do not measure forecast accuracy, stockouts, or delivery reliability.

8.4 Growth Differences

Table 4 Derived Walmart China Growth Differences

Quarter	Net minus comparable (pp)	eCommerce minus net (pp)
Q1	2.8	25.7
Q2	4.5	22.3
Q3	6.7	12.7
Q4	4.7	-0.3

Note: Author calculations from Tables 2-3. pp = percentage points. Differences use published, rounded rates.

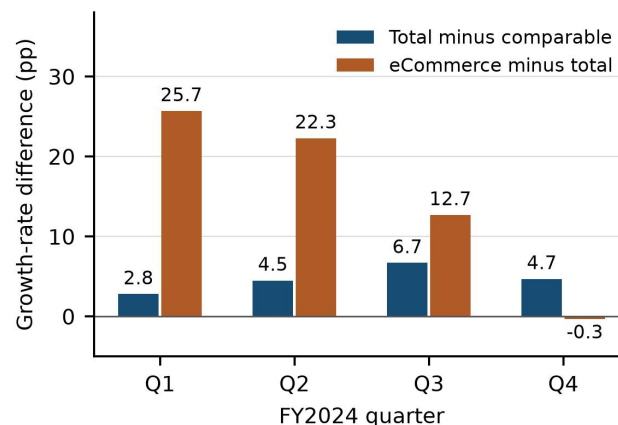


Figure 3 Growth-Rate Differences for Walmart China.

Note: Author calculations from [12-15].

Each difference equals the first growth rate minus the second. eCommerce growth exceeded total growth in Q1-Q3, then was approximately equal in Q4. The net-versus-comparable gap is not a direct estimate of new-club contribution: the measures cover different sales bases. These comparisons identify questions for operational research without attributing aggregate outcomes to a single capability.

9 DISCUSSION

A common assumption is that retail success depends mainly on products, pricing, and marketing. The analysis of Sam's Club suggests a different view, because many customer benefits are created long before consumers enter a store. Customers experience product availability, consistent quality, and competitive pricing, yet these outcomes are largely determined by upstream decisions about procurement, forecasting, inventory management, and logistics.

The four capabilities also reinforce one another rather than acting in isolation. A limited assortment makes demand easier to forecast, membership data sharpens that forecast, centralized procurement turns predictable demand into scale economies, and data-driven operations convert the resulting information into inventory decisions. This complementarity is important, because a capability that is easy to copy in isolation becomes far harder to replicate when it is embedded in a reinforcing system. The most general lesson is that modern competition increasingly occurs behind the scenes. Consumers see products, managers see supply chains, and supply chain decisions often determine which companies ultimately succeed.

10 CONCLUSION

This paper examined the factors that make the supply chain of Sam's Club successful in China. Through a limited SKU strategy, membership-driven demand visibility, centralized procurement, and data-driven operations, the company creates operational advantages that support customer value. Its success shows that supply chain management is not simply a support function but can become a powerful source of competitive advantage.

More broadly, the study suggests that modern business success depends not only on what companies sell but also on how they coordinate information, inventory, suppliers, and logistics. The analysis combines a qualitative case study with descriptive Walmart China data. The aggregate figures cannot isolate Sam's Club China or measure the precise contribution of each capability. Future work could compare Sam's Club with competing warehouse and membership retailers, collect operational data on assortment size, forecast accuracy, and inventory turnover, and test how each capability contributes to performance. Such work would clarify how firms make decisions under uncertainty and how supply chain management shapes economic outcomes. Ultimately, Sam's Club illustrates a broader lesson, namely that modern competition is increasingly a competition between supply chains rather than between products alone.

COMPETING INTERESTS

The authors have no relevant financial or non-financial interests to disclose.

REFERENCES

- [1] Fisher M L. What is the right supply chain for your product? *Harvard Business Review*, 1997, 75(2): 105–116.
- [2] Porter M E. *Competitive Advantage: Creating and Sustaining Superior Performance*. New York, NY, USA: Free Press, 1985.
- [3] Chopra S, Meindl P. *Supply Chain Management: Strategy, Planning, and Operation*. 6th ed. Boston, MA, USA: Pearson, 2016.
- [4] Choi T Y, Hong Y. Unveiling the structure of supply networks: Case studies in Honda, Acura, and DaimlerChrysler. *Journal of Operations Management*, 2002, 20(5): 469–493.
- [5] Iyengar S S, Lepper M R. When choice is demotivating: Can one desire too much of a good thing? *Journal of Personality and Social Psychology*, 2000, 79(6): 995–1006.
- [6] Lee H L, Padmanabhan V, Whang S. Information distortion in a supply chain: The bullwhip effect. *Management Science*, 1997, 43(4): 546–558.
- [7] Cachon G P, Fisher M. Supply chain inventory management and the value of shared information. *Management Science*, 2000, 46(8): 1032–1048.
- [8] Lee H L, Whang S. Information sharing in a supply chain. *International Journal of Technology Management*, 2000, 20(3–4): 373–387.
- [9] Scherer F M, Ross D. *Industrial Market Structure and Economic Performance*. 3rd ed. Boston, MA, USA: Houghton Mifflin, 1990.
- [10] Closs D J, Speier C, Meacham N. Sustainability to support end-to-end value chains: The role of supply chain management. *Journal of the Academy of Marketing Science*, 2011, 39(1): 101–116.
- [11] Sheffi Y. *The Resilient Enterprise: Overcoming Vulnerability for Competitive Advantage*. Cambridge, MA, USA: MIT Press, 2005.
- [12] Walmart Inc. FY24 Q1 Earnings Presentation. China, slide 18, May 18, 2023. <https://stock.walmart.com/sec-filings/all-sec-filings/content/0000104169-23-000043/earningspresentationfy24.htm>
- [13] Walmart Inc. FY24 Q2 Earnings Presentation. China, slide 18, Aug. 17, 2023. <https://stock.walmart.com/sec-filings/all-sec-filings/content/0000104169-23-000088/earningspresentationfy24.htm>
- [14] Walmart Inc. FY24 Q3 Earnings Presentation. China, slide 18, Nov. 16, 2023. <https://stock.walmart.com/sec-filings/all-sec-filings/content/0000104169-23-000124/earningspresentationfy24.htm>
- [15] Walmart Inc. FY24 Q4 Earnings Presentation. China, slide 18, Feb. 20, 2024. <https://stock.walmart.com/sec-filings/all-sec-filings/content/0000104169-24-000019/earningspresentationfy24.htm>